

Circular Economy Ontology Network (CEON) - Material Module

Metadata

IRI

<http://w3id.org/CEON/ontology/material/>

Title

Circular Economy Ontology Network (CEON) - Material Module

Publisher

Onto-DESIDE

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Version Iri

<http://w3id.org/CEON/ontology/material/0.4/>

Version Info

0.4

Prior Version

0.3

Preferred Namespace Prefix

ceon-material

Preferred Namespace Uri

<http://w3id.org/CEON/ontology/material/>

Description

The Material module of CEON (Circular Economy Ontology Network).

Covers Requirements

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESIDE D3.1: CVN-Resource-2, CVN-ResourceType-4, C3-3, E1-3, E2-4, E5-2, T1-1, T10-2.

In addition to requirements covered by imported ODPs, covers the following requirements from Onto-DESIDE D3.2: CE5-2, C3-3, E1-1, E1-3, E1-11, E1-13, E2-4, E3-2, E3-5, E5-2, E5-4, E5-5, T1-1, T9-2, T10-2, T10-4, T22-2.

Classes

Aluminum^C

IRI <http://w3id.org/CE0N/ontology/material/Aluminum>

Sub Class Of [ChemicalElement^C](#)

Boron^C

IRI <http://w3id.org/CE0N/ontology/material/Boron>

Sub Class Of [ChemicalElement^C](#)

Cellulose^C

IRI <http://w3id.org/CE0N/ontology/material/Cellulose>

Sub Class Of [Material^C](#)

Cerium^C

IRI <http://w3id.org/CE0N/ontology/material/Cerium>

Sub Class Of [RareEarthElement^C](#)

Chemical Element^C

IRI <http://w3id.org/CE0N/ontology/material/ChemicalElement>

Sub Class Of [ChemicalSubstance^C](#)

Super Class Of

- [Aluminum^C](#)
- [Boron^C](#)
- [Chromium^C](#)
- [Copper^C](#)
- [Iron^C](#)
- [Magnesium^C](#)
- [Manganese^C](#)
- [Nickel^C](#)
- [Niobium^C](#)
- [RareEarthElement^C](#)
- [Silicon^C](#)
- [Titanium^C](#)
- [Zinc^C](#)

Chemical Entity^C

IRI <http://w3id.org/CE0N/ontology/material/ChemicalEntity>

Sub Class Of [resourceODP:Matter^C](#)

In Range Of [hasChemicalEntity^{op}](#)

Super Class Of
[ChemicalSubstance^C](#)
[MolecularEntity^C](#)

Chemical Substance^C

IRI <http://w3id.org/CE0N/ontology/material/ChemicalSubstance>

Sub Class Of [ChemicalEntity^C](#)

Super Class Of [ChemicalElement^C](#)

Chromium^C

IRI <http://w3id.org/CE0N/ontology/material/Chromium>

Sub Class Of [ChemicalElement^C](#)

Copper^C

IRI <http://w3id.org/CE0N/ontology/material/Copper>

Sub Class Of [ChemicalElement^C](#)

Dysprosium^C

IRI <http://w3id.org/CE0N/ontology/material/Dysprosium>

Sub Class Of [RareEarthElement^C](#)

Erbium^C

IRI <http://w3id.org/CE0N/ontology/material/Erbium>

Sub Class Of [RareEarthElement^C](#)

Europium^C

IRI <http://w3id.org/CE0N/ontology/material/Europium>

Sub Class Of [RareEarthElement^C](#)

Gadolinium^C

IRI <http://w3id.org/CE0N/ontology/material/Gadolinium>

Sub Class Of [RareEarthElement^C](#)

Holmium^C

IRI <http://w3id.org/CE0N/ontology/material/Holmium>

Sub Class Of [RareEarthElement^C](#)

Iron^C

IRI <http://w3id.org/CE0N/ontology/material/Iron>

Sub Class Of [ChemicalElement^C](#)

Lanthanum^C

IRI <http://w3id.org/CE0N/ontology/material/Lanthanum>

Sub Class Of [RareEarthElement^C](#)

Lutetium^C

IRI <http://w3id.org/CE0N/ontology/material/Lutetium>

Sub Class Of [RareEarthElement^C](#)

Magnesium^C

IRI <http://w3id.org/CE0N/ontology/material/Magnesium>

Sub Class Of [ChemicalElement^C](#)

Manganese^C

IRI <http://w3id.org/CE0N/ontology/material/Manganese>

Sub Class Of [ChemicalElement^C](#)

Material^C

IRI <http://w3id.org/CE0N/ontology/material/Material>

Sub Class Of [resourceODP:Matter^C](#)

In Domain Of [hasChemicalEntity^{op}](#)
[hasMaterialComponent^{op}](#)

Restriction [hasChemicalEntity^{op}](#) some [Material^C](#)

Super Class Of [Cellulose^C](#)

Material Component^C

IRI <http://w3id.org/CE0N/ontology/material/MaterialComponent>

Sub Class Of [resourceODP:Constituent^C](#)

In Range Of [hasMaterialComponent^{op}](#)

Molecular Entity^C

IRI <http://w3id.org/CE0N/ontology/material/MolecularEntity>

Sub Class Of [ChemicalEntity^C](#)

Neodymium^C

IRI <http://w3id.org/CE0N/ontology/material/Neodymium>

Sub Class Of [RareEarthElement^C](#)

Nickel^C

IRI <http://w3id.org/CE0N/ontology/material/Nickel>

Sub Class Of [ChemicalElement^C](#)

Niobium^C

IRI <http://w3id.org/CE0N/ontology/material/Niobium>

Sub Class Of [ChemicalElement^C](#)

Praseodymium^C

IRI <http://w3id.org/CE0N/ontology/material/Praseodymium>

Sub Class Of [RareEarthElement^C](#)

Promethium^C

IRI <http://w3id.org/CE0N/ontology/material/Promethium>

Sub Class Of [RareEarthElement^C](#)

Rare Earth Element^C

IRI <http://w3id.org/CE0N/ontology/material/RareEarthElement>

Sub Class Of [ChemicalElement^C](#)

Super Class Of

[Cerium^C](#)
[Dysprosium^C](#)
[Erbium^C](#)
[Europium^C](#)
[Gadolinium^C](#)
[Holmium^C](#)
[Lanthanum^C](#)
[Lutetium^C](#)
[Neodymium^C](#)
[Praseodymium^C](#)
[Promethium^C](#)
[Samarium^C](#)
[Scandium^C](#)
[Tantalum^C](#)
[Terbium^C](#)
[Thulium^C](#)
[Ytterbium^C](#)
[Yttrium^C](#)

Samarium^C

IRI <http://w3id.org/CE0N/ontology/material/Samarium>

Sub Class Of [RareEarthElement^C](#)

Scandium^C

IRI <http://w3id.org/CE0N/ontology/material/Scandium>

Sub Class Of [RareEarthElement^C](#)

Silicon^C

IRI <http://w3id.org/CE0N/ontology/material/Silicon>

Sub Class Of [ChemicalElement^C](#)

Tantalum^C

IRI <http://w3id.org/CE0N/ontology/material/Tantalum>

Sub Class Of [RareEarthElement^C](#)

Terbium^C

IRI <http://w3id.org/CE0N/ontology/material/Terbium>

Sub Class Of [RareEarthElement^C](#)

Thulium^C

IRI <http://w3id.org/CE0N/ontology/material/Thulium>

Sub Class Of [RareEarthElement^C](#)

Titanium^C

IRI <http://w3id.org/CE0N/ontology/material/Titanium>

Sub Class Of [ChemicalElement^C](#)

Ytterbium^C

IRI <http://w3id.org/CE0N/ontology/material/Ytterbium>

Sub Class Of [RareEarthElement^C](#)

Yttrium^C

IRI <http://w3id.org/CE0N/ontology/material/Yttrium>

Sub Class Of [RareEarthElement^C](#)

Zinc^C

IRI <http://w3id.org/CE0N/ontology/material/Zinc>

Sub Class Of [ChemicalElement^C](#)

Constituent^C

IRI <http://w3id.org/CE0N/ontology/resourceODP/Constituent>

Super Class Of [MaterialComponent^C](#)

Matter^C

IRI <http://w3id.org/CE0N/ontology/resource0DP/Matter>

Super Class Of
[ChemicalEntity^C](#)
[Material^C](#)

Object Properties

has chemical entity^{op}

IRI <http://w3id.org/CE0N/ontology/material/hasChemicalEntity>

Domain [Material^C](#)

Range [ChemicalEntity^C](#)

has material component^{op}

IRI <http://w3id.org/CE0N/ontology/material/hasMaterialComponent>

Domain [Material^C](#)

Range [MaterialComponent^C](#)

Datatype Properties

anonymous formula^{dp}

IRI <http://w3id.org/CE0N/ontology/material/anonymousFormula>

descriptive formula^{dp}

IRI <http://w3id.org/CE0N/ontology/material/descriptiveFormula>

hill formula^{dp}

IRI <http://w3id.org/CE0N/ontology/material/hillFormula>

reduced chemical formula^{dp}

IRI <http://w3id.org/CE0N/ontology/material/reducedChemicalFormula>

Namespaces

:

<http://w3id.org/CEON/ontology/material/>

dcterms

<http://purl.org/dc/terms/>

odp

<http://www.ontologydesignpatterns.org/schemas/cpannotationschema.owl#>

owl

<http://www.w3.org/2002/07/owl#>

prov

<http://www.w3.org/ns/prov#>

rdf

<http://www.w3.org/1999/02/22-rdf-syntax-ns#>

rdfs

<http://www.w3.org/2000/01/rdf-schema#>

resourceODP

<http://w3id.org/CEON/ontology/resourceODP/>

vann

<http://purl.org/vocab/vann/>

Legend

c	Classes
op	Object Properties
dp	Datatype Properties